

EN 603: Energy Storage Technologies

Course Code	EN 603
Course Name	Energy Storage Technologies
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	M.Tech. in Energy Engg., MS, PhD, Elective for Final year UG
Semester	odd/even
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Preamble

The emerging energy generation sources such as solar and wind generates energy in variable patterns. Hence, energy storage is becoming of major importance to store and supply energy without any interruption. The energy storage can be in mechanical, electrochemical, or chemical forms. This course intends to provide a basic understanding of energy storage systems, and an overview of energy storage opportunities and challenges.

Course Outline

This course will cover the performance assessment and suitability of various energy storage technologies according to specific applications.

Course Modules

- **Module 1:** Energy storage systems overview - Scope of energy storage, needs and opportunities in energy storage, Technology overview and key disciplines, and applications. (3 L)
- **Module 2:** Mechanical and thermal storage system - Introduction to major forms of mechanical storages such as flywheels, pumped hydro storage, compressed air, hydraulic accumulator, heat pumps, heat engine flywheel, hot water storage tank, vacuum solar thermal collector, steam accumulator, application of phase change materials for heat storage, characteristics of mechanical systems, efficiencies and economic evaluation of mechanical energy storage systems. (8 L)
- **Module 3:** Electrochemical storage system- Introduction to the fundamental aspects of electrochemistry, Battery working principle, primary and secondary (flow) batteries, battery performance evaluation methods, major battery chemistries and their voltages; supercapacitors working principle, types of supercapacitors, cycling and performance characteristics, difference between battery and supercapacitors, Introduction to Hybrid electrochemical supercapacitors; Stand alone and grid tied energy storage systems. (11 L)
- **Module 4:** Chemical storage system- hydrogen, synthetic natural gas, biofuels and biomass, liquid nitrogen, concept of chemical storage of solar energy, application of chemical energy storage system, advantages and limitations of chemical energy storage, challenges and future prospects of chemical storage systems. (10 L)
- **Module 5:** Electromagnetic storage systems - double layer capacitors with electrostatically charge storage, superconducting magnetic energy storage (SMES), concepts, advantages and limitations of electromagnetic energy storage systems, and future prospects of electrochemical storage systems. (10 L)

Text Books

1. Frank S. Barnes and Jonah G. Levine, *Large Energy Storage Systems Handbook (Mechanical and Aerospace Engineering Series)*, CRC press (2011)
2. Robert A. Huggins, *Energy storage*, Springer Science & Business Media (2010)
3. Ralph Zito, *Energy storage: A new approach*, Wiley (2010)